

Static Load Estimation for Wheel Loader

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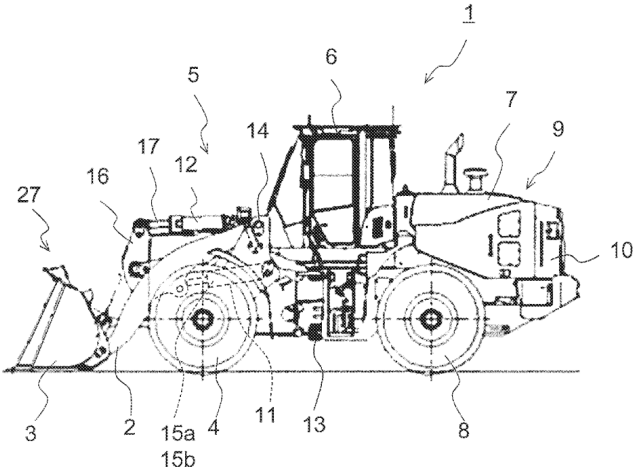
Project Overview

Environment

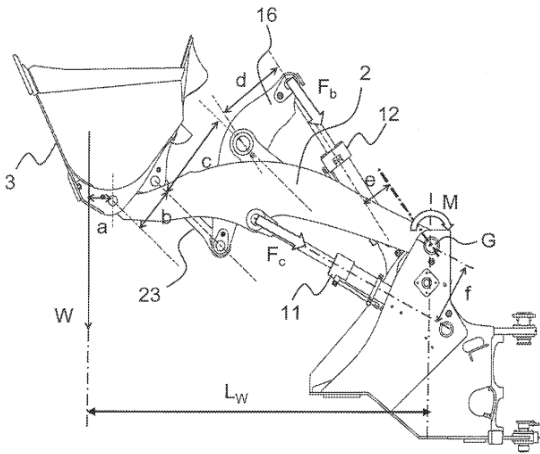
- OS: ubuntu 24.04
- MATLAB 2025b

// ### File Structure

Wheel Loader Model



Overall model



Detail model

Variables

Load Calculation Variables

Variable	Description	Unit
n	number of cylinders	-
Ab	bottom side pressure receiving area of cylinder	cm ²
Pb	bottom pressure of cylinder	MPa
Ar	rod side pressure receiving area of cylinder	cm ²
Pr	rod pressure of cylinder	MPa
M	moment around lift arm hinge pin	Nm
Fc	force applied to lift arm cylinder	N
Fb	force applied to bucket cylinder	N

Variable	Description	Unit
f	horizontal length between hinge pin and vector of F_c	m
e	horizontal length between hinge pin and vector of F_b	m
a	horizontal length between load center and bucket pin hinge pin	m
b	distance between bucket pins	m
c	distance between push rod center length and bell crank center pin	m
d	distance between bell crank pins	m
M1	moment around lift arm hinge pin in loaded state	Nm
M0	moment around lift arm hinge pin in unloaded state	Nm
W	load	kg
Lw	horizontal length from gravity center position of load	-

Dimension & Linkage Posture Variables

Variable	Description	Unit
θ_g	Lift arm angle $\angle$ (horizontal, L_{ag})	deg
Ldf	Length between bell crank D pin and bucket cylinder root F pin	mm
Ldg	Length between bell crank D pin and lift arm hinge G pin	mm
Lfg	Length between bucket cylinder root F pin and lift arm hinge G pin	mm
LfgX	Horizontal length between bucket cylinder root F pin and lift arm hinge G pin	mm
LfgY	Vertical length between bucket cylinder root F pin and lift arm hinge G pin	mm
$\angle DGA$	$\angle(L_{dg}, L_{ag})$	deg
Laf	Length between lift arm tip A pin and bucket cylinder root F pin	mm
Lag	Length between lift arm tip A pin and hinge G pin	mm
$\angle FGO$	$\angle(L_{fg}, \text{Horizontal})$	deg
θ_f	Bucket cylinder posture angle $\angle$ (horizontal, L_{ef})	deg
Lef	Bucket cylinder stroke length	mm
θ_e	Bell crank posture angle $\angle(L_{de}, L_{ef})$	deg
Lde	Bell crank DE pin length	mm
$\angle ADC$	$\angle(L_{ad}, L_{cd})$	deg
Lad	Length between lift arm tip A pin and center D pin	mm
θ_d	$\angle(L_{de}, L_{cd})$	deg
Lac	Length between lift arm tip A pin and center C pin	mm
Lcd	Bell crank CD pin length	mm
θ_c	$\angle(L_{cd}, L_{bc})$	deg

Variable	Description	Unit
Lbc	Distance between push rod BC pins	mm
Lab	Distance between bucket AB pins	mm
θb	∠(Lbc, Lab)	deg
LloadG	load center length	-
HloadG	load center angle	-
HbmcyL	lift arm cylinder angle	-

Equations

Phase 1: Link Posture Angles & Base Distances

These formulas rely on the law of cosines and geometric constants to establish the fundamental angles of the linkage based on the lift arm angle θ_g .

Sequence	Doc Formula	Equation	Description / Purpose
1	(15)	$L_{df} = \sqrt{L_{dg}^2 + L_{fg}^2 - 2L_{dg}\{L_{fgX} \cos(\theta_g + \angle DGA) + L_{fgY} \sin(\theta_g + \angle DGA)\}}$	Calculates the distance between bell crank D pin and bucket cylinder root F pin.
2	(16)	$L_{af} = \sqrt{L_{ag}^2 + L_{fg}^2 - 2L_{ag}L_{fg} \cos(\theta_g - \angle FGO)}$	Calculates the distance between lift arm tip A pin and bucket cylinder root F pin.
3	(17)	$\theta_f = \tan^{-1}\left\{\frac{L_{dg} \sin(\theta_g + \angle DGA) - L_{fgY}}{L_{dg} \cos(\theta_g + \angle DGA) - L_{fgX}}\right\} + \cos^{-1}\left(\frac{L_{df}^2 + L_{de}^2 - L_{ef}^2}{2L_{df}L_{de}}\right)$	Derives the bucket cylinder posture angle.
4	(18)	$\theta_e = \cos^{-1}\left(\frac{L_{de}^2 + L_{ef}^2 - L_{df}^2}{2L_{de}L_{ef}}\right) - 180$	Derives the bell crank posture angle.
5	(19)	$\angle ADC = \cos^{-1}\left(\frac{L_{ad}^2 + L_{dg}^2 - L_{ag}^2}{2L_{ad}L_{dg}}\right) + \cos^{-1}\left(\frac{L_{df}^2 + L_{de}^2 - L_{ef}^2}{2L_{df}L_{de}}\right) - 180 - \theta_d$	Derives the angle

Sequence	Doc Formula	Equation	Description / Purpose
			$\angle ADC$.
6	(20)	$L_{ac} = \sqrt{L_{ad}^2 + L_{cd}^2 - 2L_{ad}L_{cd} \cos \angle ADC}$	Calculates the distance between lift arm tip A pin and center C pin.
7	(21)	$\theta_c = 180 + \cos^{-1}\left(\frac{L_{ac}^2 + L_{bc}^2 - L_{ab}^2}{2L_{ac}L_{bc}}\right) - \cos^{-1}\left(\frac{L_{cd}^2 + L_{ac}^2 - L_{ad}^2}{2L_{cd}L_{ac}}\right)$	Derives the link angle θ_c .
8	(22)	$\theta_b = \cos^{-1}\left(\frac{L_{ab}^2 + L_{bc}^2 - L_{ac}^2}{2L_{ab}L_{bc}}\right) - 180$	Derives the link angle θ_b .
9	(14)	$\theta_i = \tan^{-1}\left\{\frac{L_{gi} \sin(\theta_g + H_{bmcyt}) + L_{ghY}}{L_{gi} \cos(\theta_g + H_{bmcyt}) + L_{ghX}}\right\} - (\theta_g + H_{bmcyt})$	Derives the posture angle required to calculate the horizontal distance f .

Phase 2: Inter-Link Distances & Load Coordinates

By substituting the variables from Phase 1 into trigonometric functions, the system determines the specific horizontal and vertical inter-link distances.

Sequence	Doc Formula	Equation	Description / Purpose
10	(7)	$a = L_{loadG} \cos(\theta_g + \theta_c + \theta_d + \theta_e - \theta_b + 180 - H_{loadG})$	Calculates the horizontal length between load center and bucket pin hinge pin.
11	(8)	$b = -L_{ab} \sin \theta_b$	Calculates distance b .
12	(9)	$c = L_{cd} \sin \theta_c$	Calculates distance c .
13	(10)	$d = -L_{de} \sin \theta_e$	Calculates distance d .
14	(11)	$e = L_{fg} \sin\{(180 - \theta_f) + \angle FGO\}$	Calculates the horizontal length between hinge pin and vector of F_b .
15	(12)	$f = L_{gi} \sin \theta_i$	Calculates the horizontal length between hinge pin and vector of F_c .
16	(13)	$L_w = L_{ag} \cos \theta_g + L_{loadG} \cos(\theta_g + \theta_c + \theta_d + \theta_e - \theta_b + 180 - H_{loadG})$	Calculates the horizontal length from gravity center position of load.

Phase 3: Force, Moment, and Final Load Calculation

Using the geometric distances from Phase 2 and the actual cylinder pressures, the system calculates the internal forces, the moments around the hinge pin, and finally the payload W .

Sequence	Doc Formula	Equation	Description / Purpose
17	(3)	$F_c = n(A_bP_b - A_rP_r)$	Calculates the force applied to the lift arm cylinder using bottom and rod pressures.
18	(4)	$F_b = W \cdot (\frac{a}{b}) \cdot (\frac{c}{d})$	Calculates the theoretical force applied to the bucket cylinder based on moment equilibrium.
19	(1)	$M = F_c f + F_b e$	Derives the general moment around the lift arm hinge pin supported by the cylinders.
20	(5)	$M_1 = 2f(A_bP_b - A_rP_r) + W(\frac{a \cdot c \cdot e}{b \cdot d})$	Expresses the moment M_1 in the loaded state by combining formulas 1, 3, and 4.
21	(2)	$W = \frac{M_1 - M_0}{L_w}$	General relational expression for calculating load W from loaded and unloaded moments.
22	(6)	$W = \frac{2f(A_bP_b - A_rP_r) - M_0}{L_w - (\frac{a \cdot c \cdot e}{b \cdot d})}$	The Final Load Calculation Formula: Determined by substituting formula 5 into formula 2 to solve for W directly from sensor values.

Reference

- **Patent:** US 2020/0131739 A1 (Hitachi Construction Machinery, Apr. 30, 2020).